

UPAY SHUROKKHA

Safer payments. Smarter agent liquidity.

AI Hackathon 2026 · DIU CPC × upay

Independent prototype · Synthetic data only

Decisions the prototype supports

For agents

Will I have enough cash and float for the next customer?

Forecast both demand streams.
Choose an affordable exchange.
Measure served demand and net costs.

For customers

What should I verify before sending?

Read the payment context.
Explain the warning signs.
Offer a safe next action and review.

Cash and float move in opposite directions

Customer cash-in: 1,000 BDT

Agent cash +1,000 | Agent float −1,000

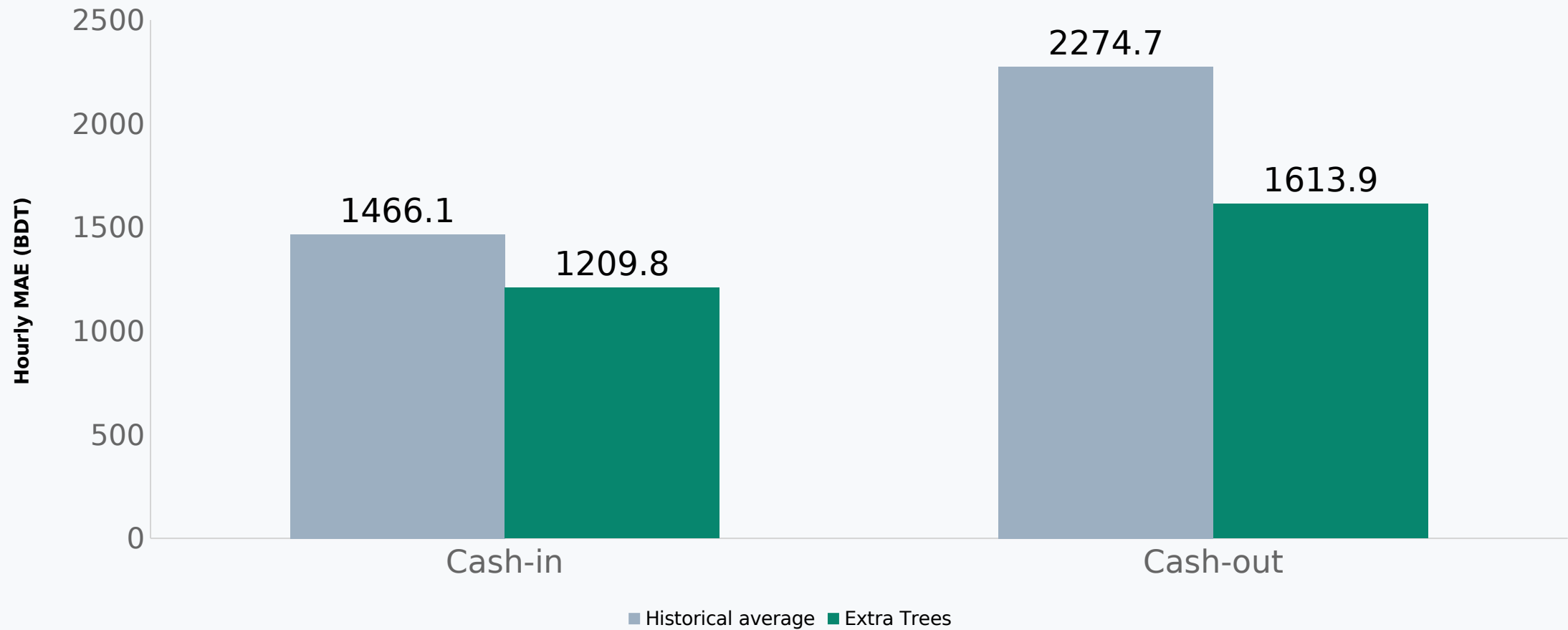
Customer cash-out: 1,000 BDT

Agent cash −1,000 | Agent float +1,000

An exchange changes the balance mix. Total working capital stays constant.

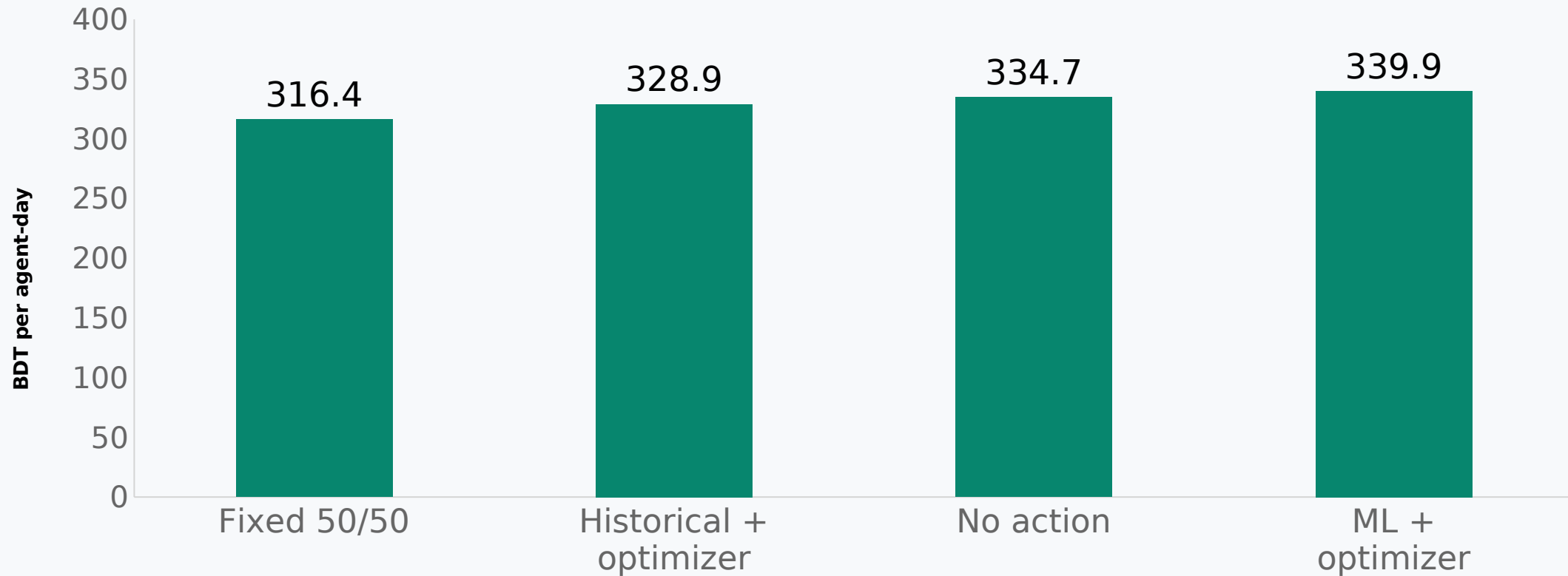
The optimizer includes no action and skips exchanges that become unaffordable.

Forecast error on unseen days



Lower is better · 3 agents × 21 test days × 12 hours · All demand is synthetic

Net earnings after exchange costs



+10.99 BDT vs historical forecasting; +5.22 BDT vs no action

Illustrative rates and capital · These outcomes do not establish commercial returns

Agent Planning: forecast, act, replay

Available working capital

₹30,000

Forecast cash-in demand

₹36,681

Forecast cash-out demand

₹71,963

Expected net change

₹35

Suggested action · Before 09:00, exchange ₹20,000 of electronic float for physical cash.

Recommendation assumes an available exchange partner and completion before the stated hour. It must be rechecked against actual balances. Future partner availability is not modeled.

Without rebalancing, the point-forecast replay first refuses a request around 15:00. This is a scenario result, not a guarantee.

Try high exchange costs, low capital and an unannounced cash-out surge.

Payment Safety: evidence before action

Last submitted check: DEMO-6630E6 · Suspicious account request · ₳8,000. Changed inputs require a new check.

Concern index / 100

98

High concern

- The description resembles scam-related examples in the small synthetic training set.
- First payment to this recipient; this alone does not imply wrongdoing.
- The amount is 8.0 times the sender's usual transfer.
- The synthetic profile contains reviewed adverse cases; an analyst should inspect the evidence.

Pause and independently verify the request.

টাকা পাঠানোর আগে স্বাধীনভাবে প্রাপকের পরিচয় ও অনুরোধ যাচাই করুন।

Cancel simulated transfer

Request human review

Continue simulation

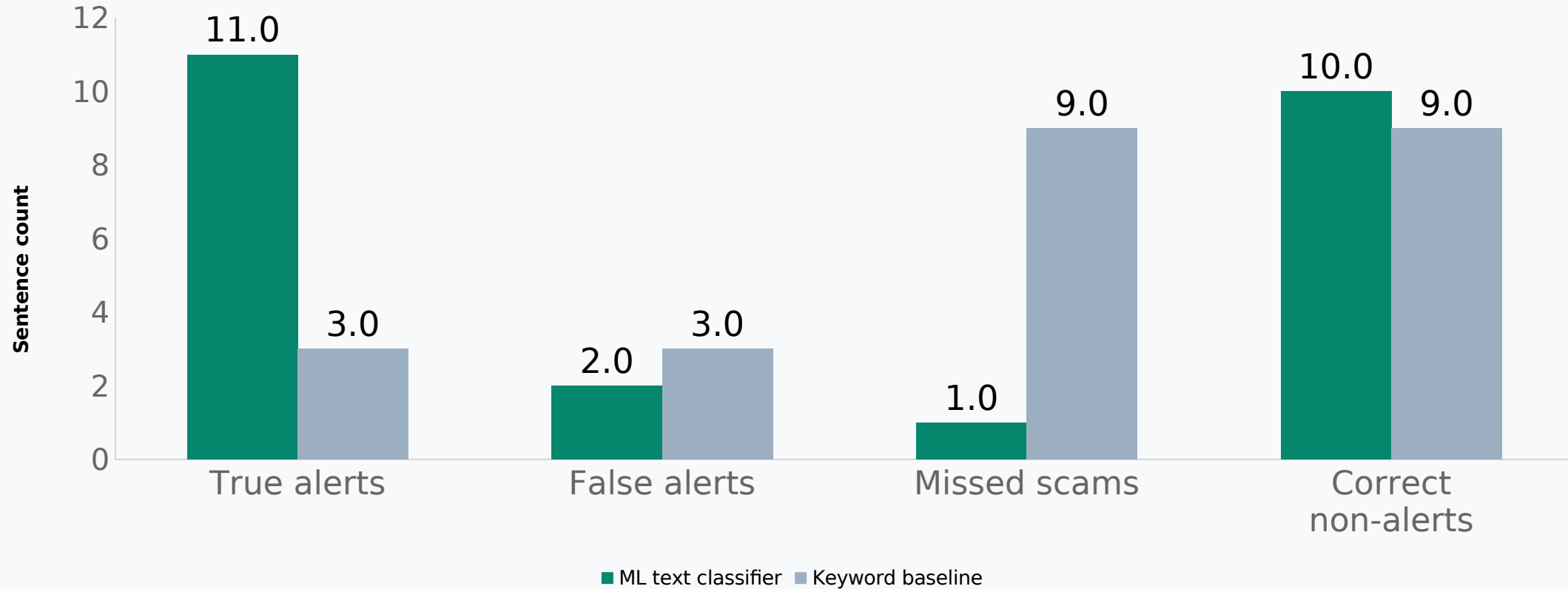
☐ I independently verified the recipient and purpose

Account/device warnings require provider-approved controls in production. This demo does not authorize payments.

› Inspect score contributions

Text classifier + explicit context rules · No real transfer, accusation or account restriction

Text classifier: useful signal, small test



84.6% precision · 91.7% recall · 16.7% false-warning rate

Only 24 test sentences, eight per language · Full concern score is not calibrated

Limits that shape the next experiment

Liquidity

Refused demand may be missing from real histories.
Partner availability, travel time and financing need modeling.

Payment safety

Negation and paraphrases can cause mistakes.
New-account and language cohorts need much larger tests.

Pilot first in shadow mode, then test human-approved actions.

Measure full net costs, missed demand, false warnings, reviewer effort and comprehension.

Working prototype and Phase 2 path

Built and reproducible

Application, synthetic data, models, policy, evaluation, 15 automated tests, report and demonstration script.

Adaptable for on-site requirements

Change capital and costs, add partner constraints, update forecasts or revise the review policy.